

Lecture 09-18

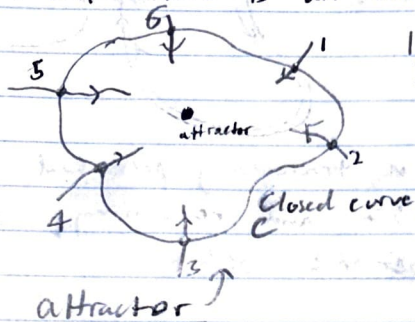
- in 1D, all you can do is "march" or stop
- in 2D, the only "new" type is a periodic orbit

Strogatz Section 8.8:

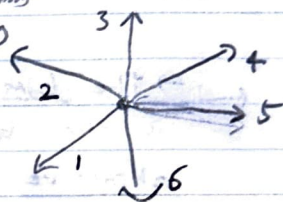
Index Theory (ideas from complex analysis)

→ not useful above 2D, but it carries topological significance

→ suppose there is an unknown equilibrium point

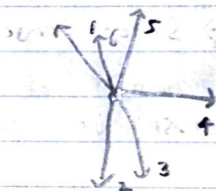
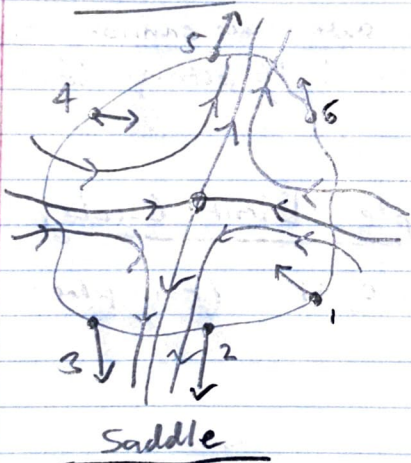


1, ..., 6: small integrations from points along curve



$$I_C = 1 \cdot 2\pi$$

(1 full rotation, $\Rightarrow$ FP inside)

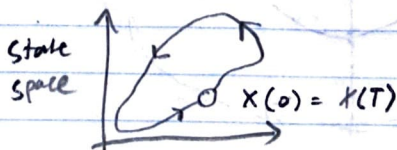


$$I_C = -1 \cdot 2\pi$$

- topological method $\Rightarrow$ not dependent on "lasso" shape
- similar to residue in Complex Analysis

7. Limit Cycles

chaosbook.org: periodic orbit

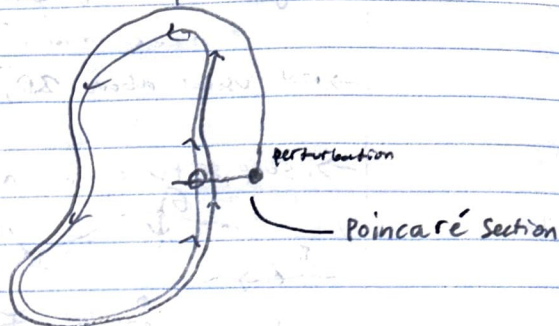
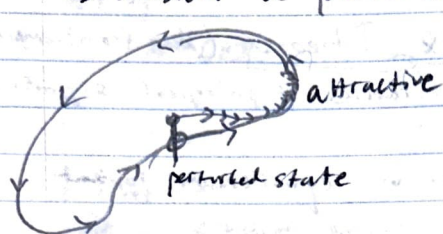


- Limit cycle = periodic orbit
- periodic orbits exist in higher dimensions

"Finkelstein: the most brilliant physicist to walk through Healey"

Limit Cycle

→ consider a perturbation from a periodic orbit;



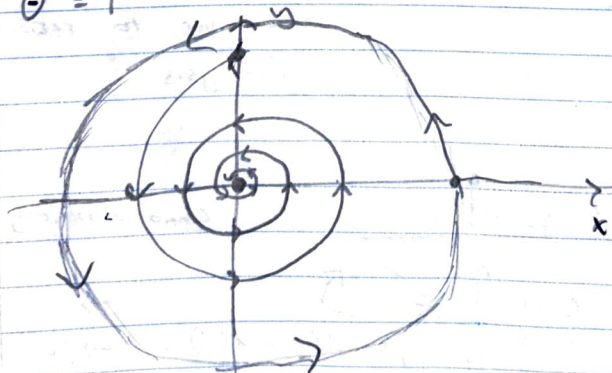
→ the Limit Cycle

is the behavior of a trajectory perturbed from an attractive periodic orbit and returning, or perturbed from a repelling periodic orbit.

→ Since trajectories in state space cannot cross, the behavior of the perturbed trajectory (distance from PO) must be monotonic.

Example 7.1.1: Simple Limit Cycle,

$$\dot{r} = r(1 - r^2) \quad r = 0, +1 \quad (r \text{ in polar: } r \in \mathbb{R}^+)$$
$$\dot{\theta} = 1$$



Example 7.1.2 Van der Pol Oscillator,

→ consider a current through a vacuum tube

$$\ddot{x} + \mu(x^2 - 1)\dot{x} + x = 0$$

negative $\rightarrow$ pump (makes signal stronger)

positive $\rightarrow$ damper

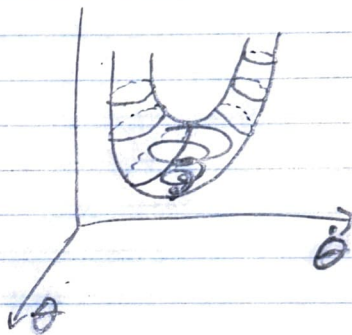
→ the dependence of the $\dot{x}$ term on the size of x^2 causes this system to result in attractive limit cycles

Damped, Quartic Oscillator

$$\ddot{\theta} + b\dot{\theta} + \sin\theta = 0$$

$$\Rightarrow \dot{x} = -\nabla V(x)$$

gradient systems



7.2 Gradient Systems,

For an n -dimensional system, if it can be written as a potential:

$$\dot{x}_i = -\nabla_i V(x)$$

→ Then, there can only be attractive fixed points

→ Lyapunov function

→ we define a $N-1$ dimensional surface to enclose a fixed point, asserting its attractiveness

→ Dulac's Criterion

→ Poincaré - Bendixson